Notes: Di Maggio, Kermani, and Palmer (RES, 2020)

How Quantitative Easing Works: Evidence on the Refinancing Channel

Contents

1 Big picture

- **Question.** How did Federal Reserve large-scale asset purchases transmit to households and the real economy through the mortgage refinancing channel?
- **Core setting.** The paper studies QE1 and later LSAPs using borrower-linked mortgage data and a segmentation design comparing conforming mortgages eligible for agency MBS purchases with jumbo mortgages that were not directly eligible.
- **Main identification idea.** QE1 purchased agency MBS, which primarily affected conforming loans below the GSE conforming loan limit. Jumbo loans above the limit provide a comparison group exposed to similar borrower and local shocks but not directly eligible for agency MBS purchases.
- **Main price result.** During QE1, conforming mortgage rates fell more than jumbo rates. The conforming-jumbo spread moved by about 40 basis points in the direction predicted by the asset-purchase channel.
- **Main quantity result.** Refinancing volumes increased much more in the conforming segment than in the jumbo segment. The preferred estimates imply a large increase in refinancing activity for QE-eligible borrowers.
- **Market segmentation result.** QE1 had larger differential effects than later QE episodes because the banking sector and jumbo securitization channel were impaired during QE1. Later programs had weaker cross-segment effects.
- **Household real effects.** Refinancing reduced monthly mortgage interest payments by roughly $250 for refinancing households and increased durable consumption, proxied by auto purchases.
- **Equity extraction and deleveraging.** The paper documents bunching around GSE loan-to-value cutoffs. Some borrowers deleveraged to become eligible, while others extracted equity through cash-out refinancing.
- **Aggregate implication.** The authors estimate that QE1 increased refinancing by about $102 billion over its first six months and raised mortgagors' consumption by roughly $13.5 billion.
- **Economic takeaway.** The real effects of unconventional monetary policy depend on the assets purchased and on market segmentation. QE works through the refinancing channel only when the purchased asset class directly affects household borrowing costs and when frictions prevent easy spillovers across segments.

2 Data and measurement

2.1 Mortgage-market data

- The paper uses Equifax Credit Risk Insight Servicing McDash (CRISM), which links Equifax credit bureau records to mortgage servicing data.
- The data contain loan terms, refinancing events, borrower credit characteristics, balances, FICO scores, loan-to-value ratios, and credit outcomes.
- The main mortgage sample focuses on first-lien, non-FHA refinance mortgages secured by single-family homes.
- For interest-rate analysis, the paper restricts to fixed-rate 15-, 20-, and 30-year mortgages with non-missing LTVs.

Interpretation: The data allow the authors to follow borrowers over time, observe refinancing directly, and link mortgage rate reductions to later household consumption behavior.

2.2 Conforming and jumbo segmentation

- Conforming loans are below the GSE conforming loan limit and are eligible for GSE guarantee and agency MBS purchase.
- Jumbo loans are above the conforming loan limit and were not directly eligible for agency MBS purchase.
- The paper compares loans above and below the conforming loan limit within county-month cells.
- The identification is strongest for QE1 because QE1 included large-scale agency MBS purchases when private jumbo securitization and banks were impaired.

Interpretation: The conforming/jumbo boundary creates a policy-relevant segmentation: two borrowers can be similar, but one segment is directly affected by agency MBS purchases while the other is not.

2.3 QE program windows

- The paper studies QE1, QE2, the Maturity Extension Program (MEP), QE3, and the 2013 tapering announcement.
- Event windows are generally the three months before and after the start of each program.
- QE1 is the main event because it included large purchases of agency MBS and occurred when the banking system was distressed.

Interpretation: Contrasting QE programs helps separate the importance of asset composition from the general macroeconomic effect of monetary stimulus.

2.4 Outcome variables

- **Interest rate:** mortgage interest rate in basis points, adjusted for borrower composition.
- **Refinancing volume:** log number or dollar volume of refinance originations in county-segment-month cells.
- **Prepayment indicator:** loan-level indicator that an existing mortgage prepays/refinances.
- **Consumption proxy:** probability that a borrower purchases a car, measured through new auto debt.
- **Equity extraction:** cash-in and cash-out refinancing measured by comparing old loan balance, new loan balance, and imputed property values.

Interpretation: The paper moves from asset prices to household-level quantities and real outcomes: rates, refinancing decisions, cash flows, and durable consumption.

3 Models and empirical specifications

3.1 Composition-adjusted interest rates

To compare mortgage rates across segments and time, the paper estimates:

$$r_{it} = \alpha_t + \beta_1(FICO_i - 760) + \beta_2(LTV_i - 0.75) + \varepsilon_{it}. \tag{4.1}$$

- r_{it} is the interest rate of loan i at time t, measured in basis points.
- The time effects α_t are rates for a representative borrower with FICO 760 and LTV 75%.
- The regression is run separately for loans above and below the conforming loan limit.

Interpretation: This adjustment ensures that observed rate movements are not driven by changes in borrower composition, such as higher FICO scores or lower LTVs later in the sample.

3.2 Interest-rate event study

The loan-level interest-rate specification is:

$$r_{icst} = \theta_0 QE_t + \theta_1 Jumbo_s + \theta_2 QE_t \cdot Jumbo_s + X_i'\beta + \gamma W_t \cdot Jumbo_s + \phi_{cs} + \eta_{ct} + \varepsilon_{icst}. \tag{4.2}$$

- $Jumbo_s$ equals one for loans above the conforming loan limit.
- θ_2 is the key coefficient: the differential response of jumbo rates relative to conforming rates after QE.
- ϕ_{cs} are county-by-segment fixed effects.
- η_{ct} are county-by-month fixed effects.

Interpretation: A positive θ_2 during QE1 means jumbo rates fell less than conforming rates, so agency MBS purchases lowered conforming rates more strongly.

3.3 Refinancing quantity regression

The quantity specification compares log refinancing volumes by county, segment, and month:

$$\log(Q_{sct}) = \psi_0 QE_t + \psi_1 Jumbo_s + \psi_2 QE_t \cdot Jumbo_s + X_{sct}'\beta + \kappa W_t \cdot Jumbo_s + \phi_{cs} + \eta_{ct} + \varepsilon_{sct}. \tag{4.3}$$

- Q_{sct} is the number or amount of refinancing originations in segment s, county c, month t.
- ψ_2 captures whether jumbo refinancing volumes respond differently than conforming volumes.
- Negative ψ_2 during QE1 means conforming refinancing increases more than jumbo refinancing.

Interpretation: This is the central quantity test. If QE1 directly lowers conforming rates, conforming refinancing should rise more than jumbo refinancing.

3.4 Loan-level prepayment regressions

The paper estimates whether individual loans prepay/refinance during QE1:

$$Prepay_i = \alpha + \delta_1 QE1_i + \delta_2 (QE1_i \times Segment_i) + X_i'\beta + FixedEffects + \varepsilon_i. \tag{1}$$

The segment interactions include jumbo status, super-jumbo status, loan amounts near the conforming loan limit, and LTV bins.

Interpretation: These specifications show which borrowers could actually access the refinancing channel. Jumbo and high-LTV borrowers were less able to refinance during QE1.

3.5 Consumption event study

For borrowers who refinanced, the event-study specification is:

$$y_{jkt} = \sum_{\tau \neq 0} \delta_\tau \mathbf{1}\{t - refinance\ month_i = \tau\} + \alpha_{kt} + \varepsilon_{jkt}. \tag{6.1}$$

- y_{jkt} is either monthly mortgage interest payment or an indicator for a new car purchase.
- α_{kt} are calendar-month by balance-quartile fixed effects.
- The omitted period is the month of refinancing.

Interpretation: The event study estimates how refinancing changes household cash flow and durable consumption before and after the refinance event.

4 Identification and empirical design

4.1 Segmented mortgage markets

- QE1 purchased agency MBS tied to conforming mortgages.
- Jumbo mortgages were not directly eligible for GSE guarantee or Fed agency MBS purchases.
- The conforming and jumbo segments had similar pre-QE trends in rates and origination volumes.
- The sharp divergence after QE1 is attributed to the Fed's asset purchases interacting with market segmentation.

Interpretation: The conforming/jumbo comparison is a difference-in-differences design around policy exposure created by the conforming loan limit.

4.2 Why QE1 is especially informative

- QE1 included large agency MBS purchases.
- The private-label jumbo securitization market had collapsed before the sample period.
- Banks were distressed and did not easily reallocate capital toward jumbo refinancing.
- This created a strong difference between the directly purchased segment and the ineligible segment.

Interpretation: The QE1 environment makes market segmentation bind. Later QE programs produced less differential response because markets and banks were less impaired or the purchases were different.

4.3 Addressing confounds

- County-month fixed effects absorb local credit demand, housing shocks, and macro shocks that affect both segments in a county.
- County-segment fixed effects absorb persistent local differences between conforming and jumbo markets.
- Loan-level controls absorb changing borrower risk composition.
- Time-series controls interacted with jumbo status absorb differences in guarantee fees, bank distress, and jumbo credit spreads.

Interpretation: The key identifying assumption is conditional parallel trends across conforming and jumbo loans within counties around short QE event windows.

4.4 Limits of the design

- Jumbo and conforming borrowers are not identical.
- Mortgage choice near the conforming loan limit can be endogenous.
- Refinancing decisions are borrower choices and may be correlated with unobserved household shocks.
- Consumption event studies are conditional on refinancing, so they document effects of refinancing rather than an entirely randomized cash-flow shock.

Interpretation: The paper combines several designs to address these concerns: segment comparisons, bank-health heterogeneity, loan-level prepayment models, LTV bunching, and borrower event studies.

5 Tables and figures

Visuals are directly extracted from the paper and grouped compactly to save space.

5.1 Figure 1 and Table 1: QE timeline and sample characteristics

Read:

- Figure 1 shows the large expansion of the Fed balance sheet from 2008 to 2015 and the timing of QE1, QE2, MEP, QE3, and tapering.
- Table 1 shows that conforming loans are much more numerous than jumbo loans: about 6.7 million conforming observations versus about 156 thousand jumbo observations.
- Conforming borrowers have average FICO around 752 and average rate around 4.54%; jumbo borrowers have average FICO around 762 and average rate around 4.09%.
- Panel III reports time-series controls, including guarantee fees, CDS spreads, and FICO credit spreads.

Economic message: The empirical strategy exploits a large policy shock in a market where conforming and jumbo loans differ in direct QE eligibility.

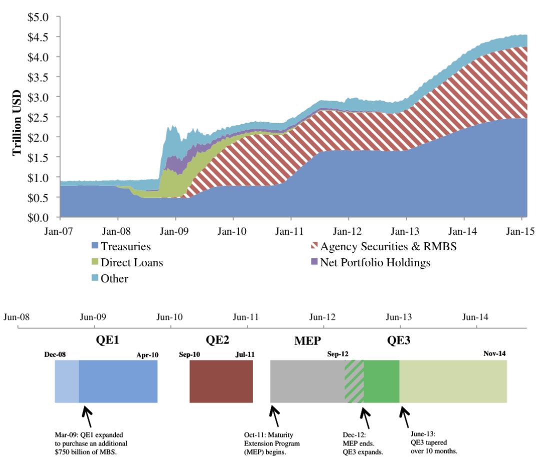

FIGURE 1

Federal reserve balance sheet and QE timeline

Figure 1: Fed balance sheet and QE timeline

TABLE 1

Summary statistics

	Mean	Std. Dev.	10th	50th	90th
Panel I. Conforming loans					
LTV (%)	65.90	22.18	36.83	68.38	88.96
FICO score	752.19	51	681	766	805
Interest rate (%)	4.54	0.97	3.38	4.50	5.75
Balance ($)	206,958	118,172	79,000	180,000	385,000
Observations			6,684,123		
Panel II. Jumbo loans					
LTV (%)	64.49	24.10	40.86	66.84	80.00
FICO score	761.59	36.22	709	770	801
Interest rate (%)	4.09	1.15	2.75	3.88	5.63
Balance	1,033,282	1,208,891	535,000	815,000	1,494,000
Observations			155,787		
Panel III. Time-series controls					
Guarantee fees (bp)	34.05	11.03	25.70	28.92	56.07
CDS spread (bp)	115.62	122.29	16.41	80.51	240.61
FICO credit Spread (bp)	6.24	3.33	1.03	6.58	9.59
Observations			72		

Notes: Panel I and Panel II report loan-level summary statistics on conforming and jumbo loans from Equifax's CRISM database which merges McDash Analytics mortgage servicing records from Lender Processing Services with Equifax credit bureau data. Panel III reports summary statistics for time series controls used in robustness checks, including guarantee fees from Fuster *et al.* (2013), CDS index spreads from Markit, and the FICO credit spread as calculated by the authors using CRISM.

Table 1: Summary statistics

5.2 Figure 2 and Table 2: interest-rate pass-through

Read:

- Figure 2 shows that conforming and jumbo refinance rates tracked each other before QE1 but diverged after QE1.
- Table 2 shows that, with controls, the QE1 $Program \times Jumbo$ coefficient is about 43.9 basis points.
- The positive coefficient means jumbo rates fell less than conforming rates during QE1.
- QE2, MEP, QE3, and tapering have much smaller or different differential effects.

Economic message: QE1 MBS purchases passed through to retail mortgage rates in the conforming segment more strongly than in the jumbo segment.

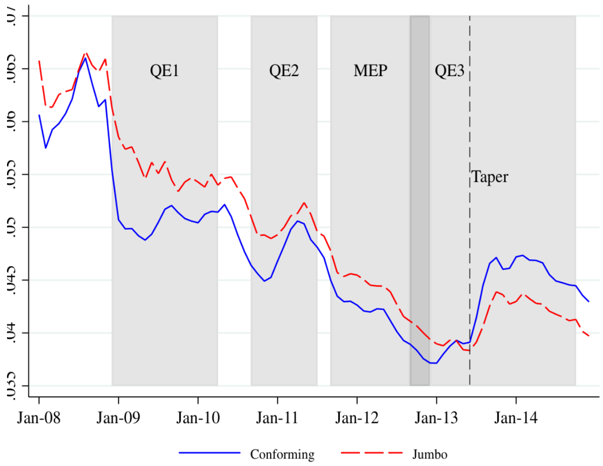

FIGURE 2

Figure 2: conforming and jumbo refinance rates

TABLE 2

Effect of QE commencement on interest rates by QE program

Program	(1) QE1	(2) QE2	(3) MEP	(4) QE3	(5) Tapering
Panel I. Without controls					
Program indicator	−120.607***	−36.271***	−47.290***	−19.874***	18.711
	(14.341)	(9.808)	(7.045)	(5.568)	(11.642)
Jumbo indicator	26.246***	45.060***	33.398***	12.668*	−4.955**
	(8.029)	(12.810)	(7.835)	(7.033)	(2.161)
Program × Jumbo	55.188**	−5.143	6.051	2.467	−14.532
	(18.762)	(13.640)	(10.457)	(7.955)	(12.765)
Observations	466,831	604,596	450,059	527,983	674,959
R-squared	0.382	0.151	0.176	0.041	0.029
Panel II. With controls					
Program × Jumbo	43.916***	−6.611*	−5.002	6.392***	−15.649**
	(5.337)	(3.187)	(2.961)	(1.648)	(5.945)
Controls:					
Loan-level	Yes	Yes	Yes	Yes	Yes
Time-series × Jumbo	Yes	Yes	Yes	Yes	Yes
County-month fixed effects	Yes	Yes	Yes	Yes	Yes
County-segment fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	466,831	604,596	450,059	527,983	674,959
R-squared	0.616	0.599	0.684	0.614	0.615

Notes: The table reports regression coefficients relating loan-level mortgage interest rates (in basis points) to indicated unconventional monetary policy programs. QE, MEP ("Maturity Extension Program"), and Tapering Indicators are dummy variables equal to one after the introduction of each program. The sample includes single-family, first-lien, 15/20/30-year term, fixed-rate, non-FHA refinance mortgages with non-missing LTVs. Jumbo Indicator is a dummy equal to one for jumbo loans. Program × Jumbo is the interaction between the program dummies and Jumbo Indicator. The event window includes the three months before/after the beginning month of each program period (*e.g.* QE1 sample is September 2008 to February 2009). Specifications in Panel II control for 5-point LTV bins, 20-point FICO bins, a categorical interaction of interest rate type, interest-only indicator, and original term, an indicator for missing FICO, county-month fixed effects, and county-segment fixed effects. The contribution of interactions of the Jumbo indicator with GSE guarantee fees, mortgage credit spreads, and bank credit default swaps on interest rates over the entire sample period were also subtracted before the Panel II specifications were run. (See Supplementary Appendix Table 1 for the coefficients of these time-series controls.) Standard errors are clustered at the month-segment level. Asterisks denote significance levels (***=1%, **=5%, *=10%).

Table 2: interest-rate effects by QE program

5.3 Figure 3 and Table 3: refinancing volumes

Read:

- Figure 3 shows a sharp rise in conforming refinance originations after QE1.
- Jumbo refinancing rises much less during the same period.
- Table 3 reports that, with controls, $Program \times Jumbo$ is about -0.810 for QE1.
- The negative coefficient means jumbo refinancing grew much less than conforming refinancing.

Economic message: Lower conforming rates translated into higher refinancing quantities. The policy affected households through both prices and quantities.

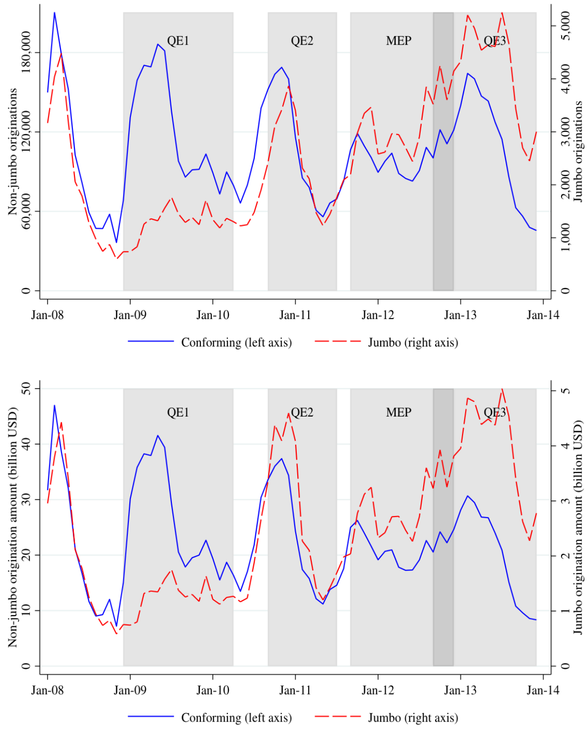

FIGURE 3

Figure 3: refinance origination volume

TABLE 3

Effect of QE commencement on log refinance origination volumes by QE program

	(1)	(2)	(3)	(4)	(5)
Program	QE1	QE2	MEP	QE3	Tapering
Panel I. Without controls					
Program indicator	1.019***	0.597***	0.544***	0.122	−0.346**
	(0.279)	(0.164)	(0.075)	(0.080)	(0.139)
Jumbo indicator	−2.138***	−2.169***	−1.757***	−1.543***	−1.435***
	(0.156)	(0.188)	(0.116)	(0.098)	(0.036)
Program × Jumbo	−0.831**	0.067	−0.057	0.060	0.416**
	(0.289)	(0.208)	(0.143)	(0.114)	(0.146)
Observations	492	492	492	492	492
R-squared	0.637	0.560	0.466	0.355	0.292
Panel II. With controls					
Program × Jumbo	−0.810***	0.073	0.231	−0.151**	0.230
	(0.197)	(0.104)	(0.154)	(0.066)	(0.157)
Controls:					
Loan composition	Yes	Yes	Yes	Yes	Yes
Time-series × Jumbo	Yes	Yes	Yes	Yes	Yes
County-month FEs	Yes	Yes	Yes	Yes	Yes
County-segment FEs	Yes	Yes	Yes	Yes	Yes
Observations	492	492	492	492	492
R-squared	0.975	0.991	0.988	0.988	0.994

Notes: Table reports regression coefficients relating county × month × mortgage segment log refinancing volumes to unconventional monetary policy programs. The left-hand side variable is the log dollar volume of refinanced mortgages at the county-month-segment level as reported in the CRISM data. QE, MEP ("Maturity Extension Program"), and Tapering Indicators are dummy variables equal to one after the introduction of each program. Jumbo Indicator is a dummy equal to one for jumbo loans. Program × Jumbo is the interaction between the program dummies and Jumbo Indicator. The sample includes single-family, first-lien, non-FHA refinance mortgages with non-missing LTV values. Counties are included in the sample if they have a positive number of jumbo originations in every sample month. The event window includes the three months before/after each QE period (*e.g.* QE1 sample is September 2008 to February 2009). Specifications in Panel II include controls for average FICO, average LTV, the fraction not missing a FICO score for all mortgages originated in that county-month-segment, county-month fixed effects, and county-segment fixed effects. The contribution of interactions of the Jumbo indicator with GSE guarantee fees, mortgage credit spreads, and bank credit default swaps on log refinancing volumes over the entire sample period were also subtracted before the Panel II specifications were run. (See Supplementary Appendix Table 1 for the coefficients of these time-series controls.) Standard errors are clustered at the month-segment level. Asterisks denote significance levels (***=1%, **=5%, *=10%).

Table 3: refinancing volume effects

5.4 Table 4 and Table 5: bank distress and prepayment access

Read:

- Table 4 shows that distressed banks originated fewer jumbo refinance loans during QE1.
- The same bank-distress interaction is weak for conforming refinance volume, consistent with GSE support insulating the conforming segment.
- Table 5 shows that QE1 increases prepayment/refinancing probability, but the effect is smaller for jumbo and high-LTV borrowers.
- Interactions with super-jumbo and high-LTV status are especially negative, showing that some borrowers could not access the refinancing channel.

Economic message: QE1 worked through the agency-conforming mortgage market. Borrowers outside the eligible segment, especially jumbo or high-LTV borrowers, were less able to refinance.

TABLE 4

Bank-level effects of QE1 and QE3 on refinance origination volumes

	(1)	(2)	(3)	(4)
Program	QE1		QE3	
Panel I. Log(jumbo volume)				
Program × bank distress	−0.667**	−0.643***	0.208	0.276
	(0.249)	(0.176)	(0.199)	(0.174)
Program indicator	0.024		0.828***	
	(0.527)		(0.152)	
Bank fixed effects	Yes	Yes	Yes	Yes
County-month fixed effects		Yes		Yes
Observations	510	510	1,314	1,314
R-squared	0.063	0.350	0.092	0.372
Panel II. Log(conforming volume)				
Program × bank distress	−0.099	−0.166	0.034	0.026
	(0.256)	(0.277)	(0.065)	(0.062)
Program indicator	0.847*		0.092*	
	(0.409)		(0.051)	
Bank fixed effects	Yes	Yes	Yes	Yes
County-month fixed effects		Yes		Yes
Observations	510	510	510	510
R-squared	0.346	0.612	0.346	0.612

Notes: Table reports regression coefficients relating a bank-level measure of bank distress to county-bank level log quantities of jumbo and conforming loans issued in QE1 and QE3. The bank distress measure is equal to the sum of net charge-offs for loans related to real-estate across all quarters in 2007 and 2008, normalized by total assets in the last quarter of 2008. Program is a dummy equal to one for the introduction of each program. The event window includes the three months before/after each QE period (*e.g.* QE1 sample is September 2008 to February 2009). Counties are included in the sample if they have a positive number of jumbo originations in every month between 2008 and 2013. Furthermore, county-bank pairs are included in the sample if they have at least 10 jumbo refinance originations in the given QE event window. Standard errors are clustered at the bank level. Asterisks denote significance levels (***=1%, **=5%, *=10%).

Table 4: bank distress and refinancing volume

TABLE 5

Loan-level estimates of QE1 effect on prepayment indicator

	(1)	(2)	(3)	(4)	(5)	(6)
QE1 indicator	0.056***	0.053***		0.058***	0.055***	
	(0.001)	(0.001)		(0.001)	(0.001)	
QE1 × Jumbo	−0.022***	−0.022***	−0.023***			
	(0.001)	(0.001)	(0.001)			
QE1 × 100–140% CLL				−0.009***	−0.009***	−0.010***
				(0.001)	(0.001)	(0.001)
QE1 × Super-jumbo				−0.059***	−0.058***	−0.063***
				(0.001)	(0.001)	(0.001)
QE1 × over 80% LTV	−0.033***	−0.027***	−0.019***			
	(0.001)	(0.001)	(0.001)			
QE1 × 80–90% LTV				−0.027***	−0.022***	−0.018***
				(0.001)	(0.001)	(0.001)
QE1 × over 90% LTV				−0.047***	−0.039***	−0.027***
				(0.001)	(0.001)	(0.001)
Jumbo indicator	−0.002***	−0.002***				
	(0.001)	(0.001)				
100–140% CLL indicator				−0.002***	−0.002***	
				(0.001)	(0.001)	
Super-jumbo indicator				−0.003***	−0.003***	
				(0.001)	(0.001)	
Over 80% LTV indicator	−0.011***	−0.012***				
	(0.000)	(0.000)				
80–90% LTV indicator				−0.009***	−0.009***	
				(0.000)	(0.000)	
Over 90% LTV indicator				−0.016***	−0.017***	
				(0.000)	(0.000)	
Constant	0.019***	0.020***		0.019***	0.020***	
	(0.000)	(0.000)		(0.000)	(0.000)	
Loan controls × QE1	Yes	Yes	Yes	Yes	Yes	Yes
Borrower controls × QE1		Yes	Yes		Yes	Yes
County-segment fixed effects			Yes			Yes
County-time fixed effects			Yes			Yes
Observations	1,219,140	1,219,140	1,219,140	1,219,140	1,219,140	1,219,140

Notes: Table reports loan-time-level linear probability model estimates of a prepayment indicator on loan and borrower characteristics and QE1 indicator interactions. Sample includes first-lien 30-year, fixed-rate mortgages with non-missing LTVs for two time periods: three months before/after QE1 announcement. Loan controls are log loan balance, current LTV bin, and non-super jumbo and super-jumbo segment indicators (current balance 100–140% and 140%+ of conforming

Table 5: prepayment and segment eligibility

5.5 Figure 4 and Figure 5: consumption and balance-sheet effects

Read:

- Figure 4 shows that refinancing lowers monthly mortgage interest payments by roughly $250 and raises the probability of new car purchases after refinancing.
- The auto-loan response suggests that households spend part of the refinancing savings on durable consumption.
- Figure 5 shows bunching at the 80% LTV cutoff, consistent with borrowers changing leverage to access GSE-eligible refinancing.
- Some borrowers cash in to deleverage toward the cutoff, while others cash out and extract home equity.

Economic message: The refinancing channel matters for real activity because it changes household cash flow and balance sheets, not just mortgage rates.

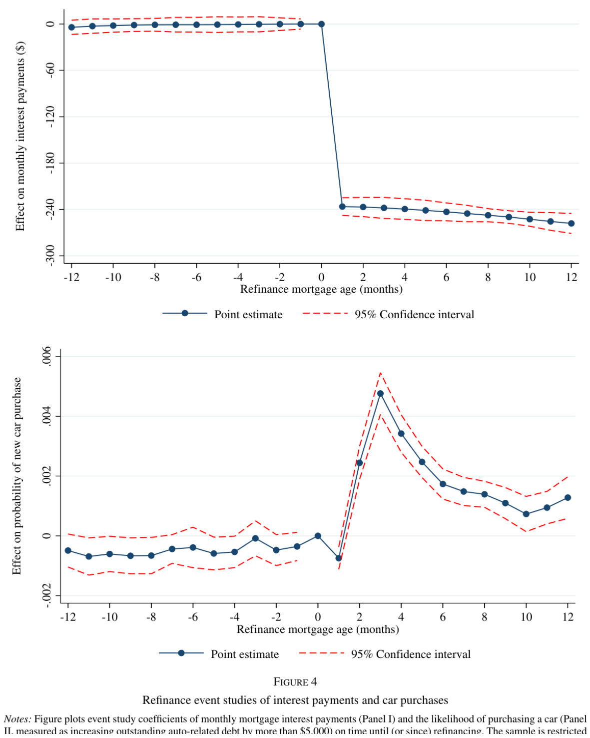

Figure 4: refinancing event studies

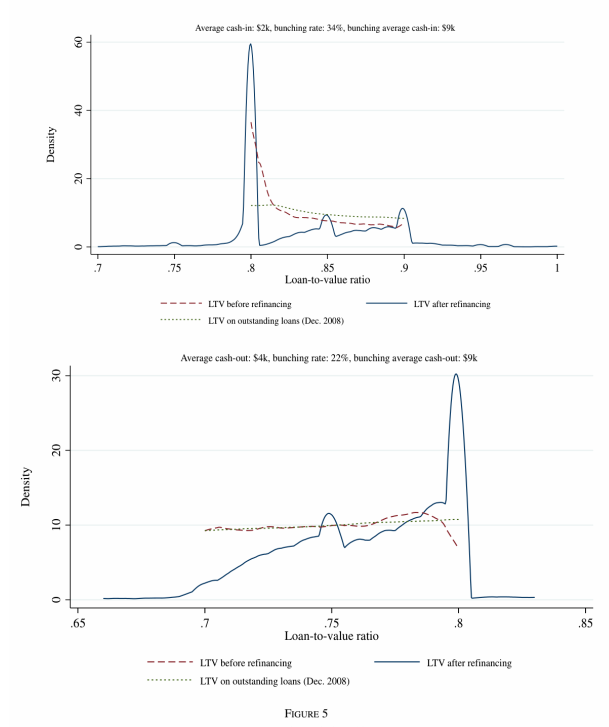

Figure 5: loan-to-value bunching

6 Short summary

- The paper studies how QE transmits through mortgage refinancing.
- QE1 agency MBS purchases lowered conforming mortgage rates relative to jumbo rates.
- The conforming segment experienced a much larger refinancing boom than the jumbo segment.
- The design uses the conforming loan limit as a segmentation boundary between QE-eligible and QE-ineligible loans.
- County-month and county-segment fixed effects help absorb local demand and supply shocks.
- Distressed banks originated fewer jumbo refinances during QE1, helping explain why spillovers to the jumbo segment were limited.
- Refinancing reduced monthly mortgage payments and increased durable consumption.
- LTV bunching shows that borrowers adjusted leverage to access refinancing, with both cash-in and cash-out behavior.
- QE transmission depends on asset composition and market segmentation.

7 Conclusion

7.1 Core conclusion

Di Maggio, Kermani, and Palmer show that QE1 worked partly through a refinancing channel. By purchasing agency MBS, the Fed reduced rates in the conforming mortgage market, increased refinancing volumes, lowered household interest payments, and stimulated durable consumption.

7.2 Economic intuition

The mechanism is simple. If the central bank buys assets tied to a specific debt market, yields in that segment fall. When households can refinance into lower rates, their monthly payments fall. This creates cash-flow relief and can increase consumption. But the effect depends on whether borrowers are eligible and whether financial intermediaries can transmit the rate decline.

7.3 Takeaway

Unconventional monetary policy is not asset-neutral. QE is more powerful when the central bank purchases assets linked to household borrowing costs in segmented markets. Its effects are weaker for households outside the purchased asset class or unable to refinance because of high LTVs, jumbo balances, or impaired intermediaries.